Nigerian Nurses' and Nurse Educators' Readiness for Technology and Artificial Intelligence in Training and Practice: A Cross-sectional Survey in Oyo State, Nigeria

Abstract

Globally, AI and technology are being integrated into nursing education and practice, providing students with realistic patient care scenarios for safe, hands-on learning. These technological advancements and AI have also aided Nurses in documentation and data input, allowing nurses to focus more on patient care. However, the adoption of AI and other technologies in Nigeria's nursing environment is still in its early stages compared to more developed countries. This study investigates the readiness of Nigerian nurses and nurse educators in Oyo state to integrate technology and Artificial Intelligence (AI) into their training and clinical practice. A quantitative survey design was employed, with 115 registered nurses from Oyo State, Nigeria, participating. The study reveals moderate levels of technological use, with 73% of respondents having used some form of technology or AI tools in their practice. Perceptions towards AI integration were predominantly positive, with over 95% agreeing on its potential to improve healthcare delivery and patient outcomes. Readiness to utilise AI was high, with almost all respondents (97.4%) ready to utilise AI tools in practice. Key barriers to

adoption included unavailability of necessary technologies (65.2%), insufficient infrastructure (62.7%), and inadequate training (60.8%). The study found significant associations between previous utilisation of AI, practice designation and readiness to utilise AI, as well as between nursing roles and perceptions of AI integration. These findings highlight the need to develop and implement comprehensive AI training programs for Nurses, focusing on hands-on experience and understanding of AI applications in patient care. Given the high level of readiness, tailored training programs can help close the gap created by inadequate training.